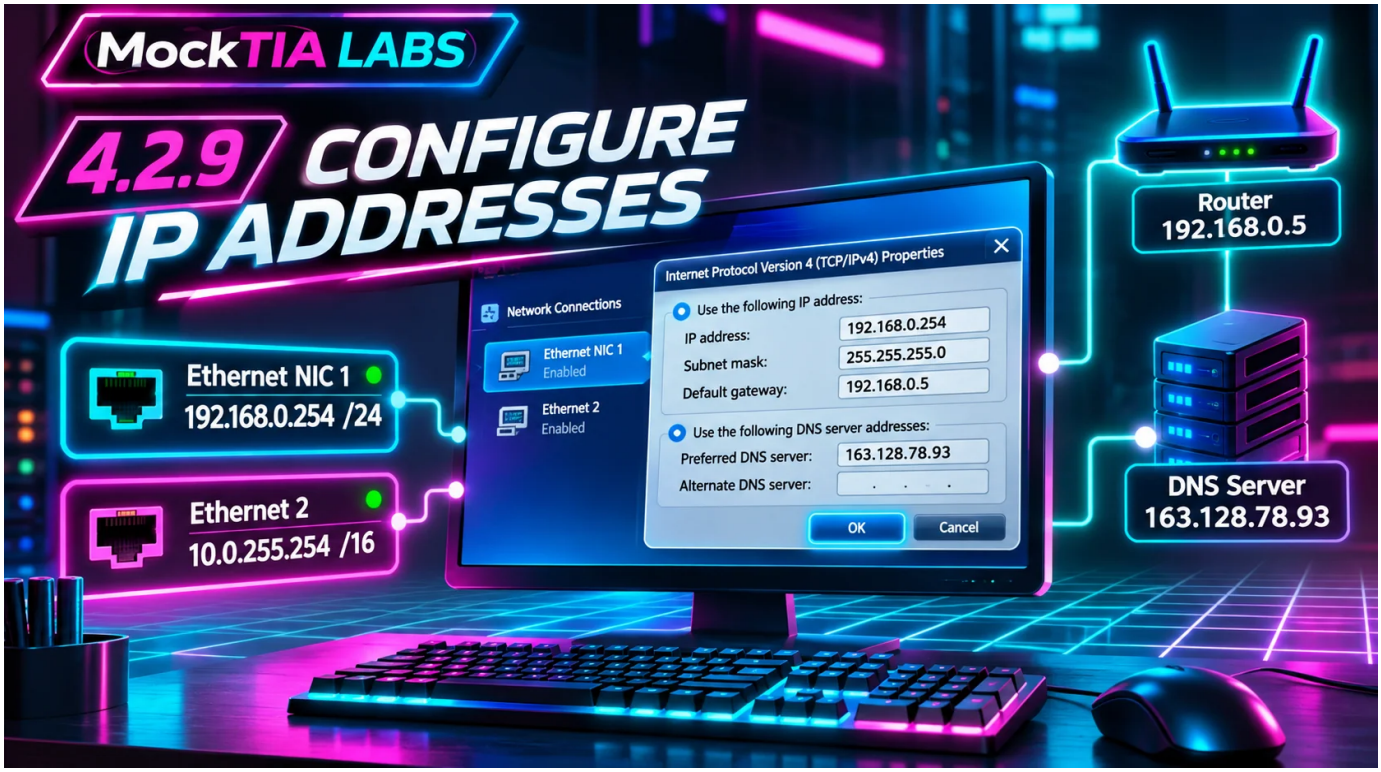
4.2.9 — Configure IP Addresses

Study Guide + Vocabulary + Illustrated Reference Diagrams



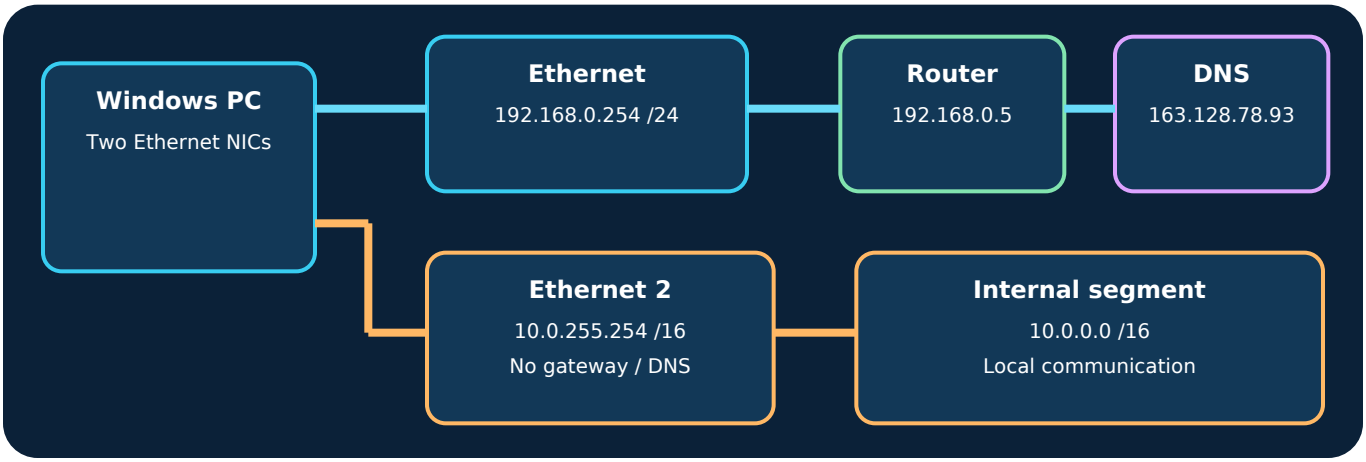
Reference image: dual-NIC Windows workstation, IPv4 settings, routing and DNS endpoints.

Learning objectives

Configure two IPv4 network adapters with different masks, explain the meaning of gateway and DNS settings, verify saved values with ipconfig /all, and use ping to validate the preferred DNS endpoint.

Scenario: An executive workstation is connected to a production network and a separate internal network. The two adapters must have distinct IP configurations. This is a standalone MockTIA training simulation based on the general 4.2.9 skill topic.

01 · Workstation Network Plan



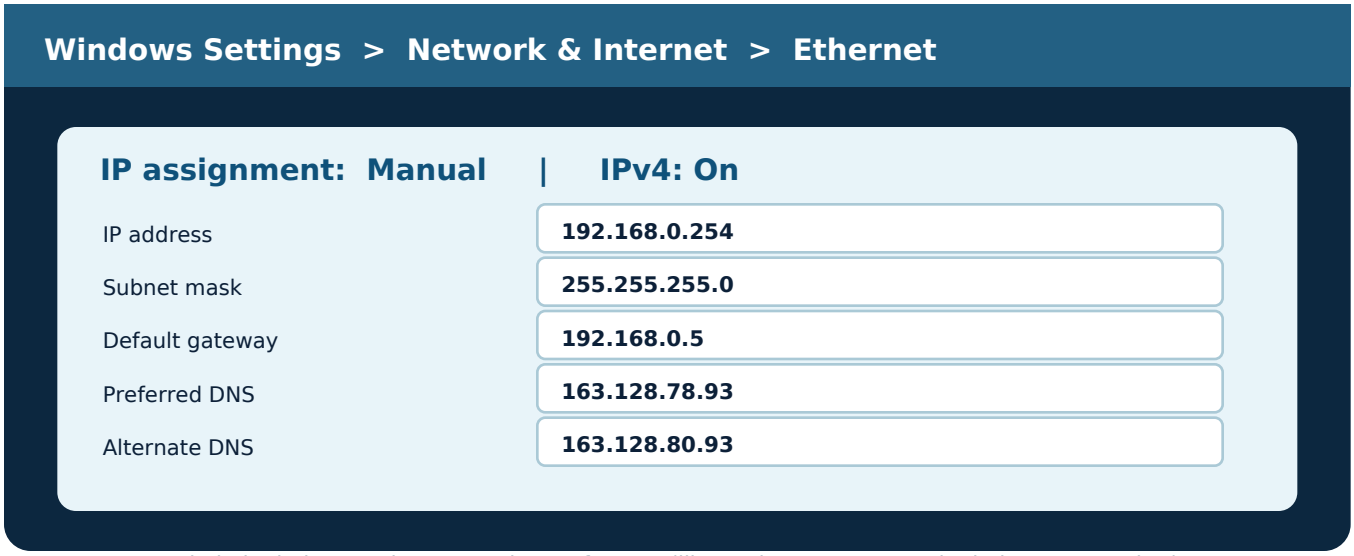
Two independently configured NICs; only the production Ethernet interface has a default gateway and DNS resolvers.

Parameter	Ethernet (production)	Ethernet 2 (internal)
Network	192.168.0.0/24	10.0.0.0/16
Static host	192.168.0.254	10.0.255.254
Subnet mask	255.255.255.0	255.255.0.0
Gateway	192.168.0.5	Leave blank
Preferred DNS	163.128.78.93	Leave blank
Alternate DNS	163.128.80.93	Leave blank

Why the addresses end in .254: In the /24 network, .0 identifies the network and .255 is broadcast, so .254 is the last usable host. In the /16 network, 10.0.0.0 is the network and 10.0.255.255 is broadcast, so 10.0.255.254 is the last usable host.

02 · Windows Configuration Reference

Ethernet — Windows Settings: Network & Internet → Ethernet → Edit IP assignment → Manual → IPv4 On. Enter the five values below and select Save.



Labeled Windows-style IPv4 settings reference (illustrative, not an actual Windows screenshot).

Ethernet 2 — Classic TCP/IPv4 properties

Open Network Connections → right-click Ethernet 2 → Properties → Internet Protocol Version 4 (TCP/IPv4) → Properties. Select manual addressing and enter 10.0.255.254 with 255.255.0.0. Leave default gateway and both DNS boxes empty. Apply and close.

Validation commands

Command	Purpose
ipconfig /all	Lists interface IPv4, masks, gateway, DNS
ping 163.128.78.93	Tests reachability of the preferred DNS server
ping 192.168.0.5	Tests reachability of the local router
route print	Displays routing information and default route

Interpretation: A successful ping demonstrates ICMP reachability to an IP address, but does not prove that DNS name resolution works. For name resolution, separately test with nslookup on a real machine.

03 · Vocabulary and Troubleshooting

Term	Meaning
IPv4 address	32-bit logical address identifying an interface.
Static IP	Address information manually assigned by a technician.
DHCP	Service that automatically leases IP settings.
Subnet mask	Separates network bits from host bits.
CIDR /24	24 network bits; mask 255.255.255.0.
CIDR /16	16 network bits; mask 255.255.0.0.
Default gateway	Router used to reach destinations beyond the local subnet.
DNS server	Resolves domain names into IP addresses.
Network address	Address representing a subnet, not an assignable host.
Broadcast address	IPv4 subnet address used for local broadcast delivery.
NIC	Network interface card/adapter.
ICMP / ping	Network-layer protocol and tool for reachability tests.
ipconfig /all	Windows command that displays detailed adapter settings.
APIPA	Windows self-assigned 169.254.0.0/16 address when DHCP fails.

Troubleshooting quick checks

Wrong mask: The host may treat remote destinations as local (or vice versa). Wrong gateway: Traffic outside the local subnet fails. Wrong DNS: Name lookup may fail even if ping to an IP succeeds. Extra gateway on Ethernet 2: Can create an unintended default route. Ping times out: Check configuration, link status, routes and ICMP filtering.

04 · Review and Student Readiness

1. What is the last valid host in 192.168.0.0/24?

_____

2. Why is 10.0.255.255 not assigned to Ethernet 2?

_____

3. What subnet mask corresponds to /16?

_____

4. What is the purpose of the default gateway on Ethernet?

_____

5. Why is the second adapter configured without DNS or a gateway?

_____

6. Does successfully pinging a DNS server prove that hostname resolution works? Explain.

_____

Answer key (for self-study)

1. 192.168.0.254. 2. It is the subnet broadcast address. 3. 255.255.0.0. 4. It forwards traffic to remote networks. 5. The work order restricts that NIC to the internal segment. 6. No. Ping by numeric IP tests reachability; DNS resolution requires a separate lookup test.

Before submitting your lab

[ ] Ethernet saved as 192.168.0.254 /24 with router 192.168.0.5 and both DNS values.

[ ] Ethernet 2 saved as 10.0.255.254 /16 with gateway and DNS blank.

[ ] PowerShell ping to preferred DNS 163.128.78.93 succeeded in the simulation.

[ ] Knowledge questions verified; student full name entered.

[ ] Used Print / Save Assessment to generate the Canvas submission PDF.

Student completion report is generated by the HTML simulation. This study guide is not the graded submission.